

AL-TAHIR Showroom - Professional Project Documentation & User Flow

1. Project Overview & Architecture

This section highlights the professional software engineering principles applied throughout the application.

Entity-Relationship Architecture (Data Flow)

The application is built on a highly relational data architecture.

- A **Purchase** automatically updates **Inventory** records.
- The **Inventory** feeds directly into **Dashboard Metrics**.
- Creating a **Sale** instantly generates an **Installment Contract** and updates **Customer History**.

Dynamic State Management

The Executive Command Center (Dashboard) utilizes dynamic state management to render real-time calculations. Any **Investment** addition or **Withdrawal** made instantly recalculates the *Capital Investment*, *Stock Allocation*, and *Performance Velocity* without requiring manual refreshes.

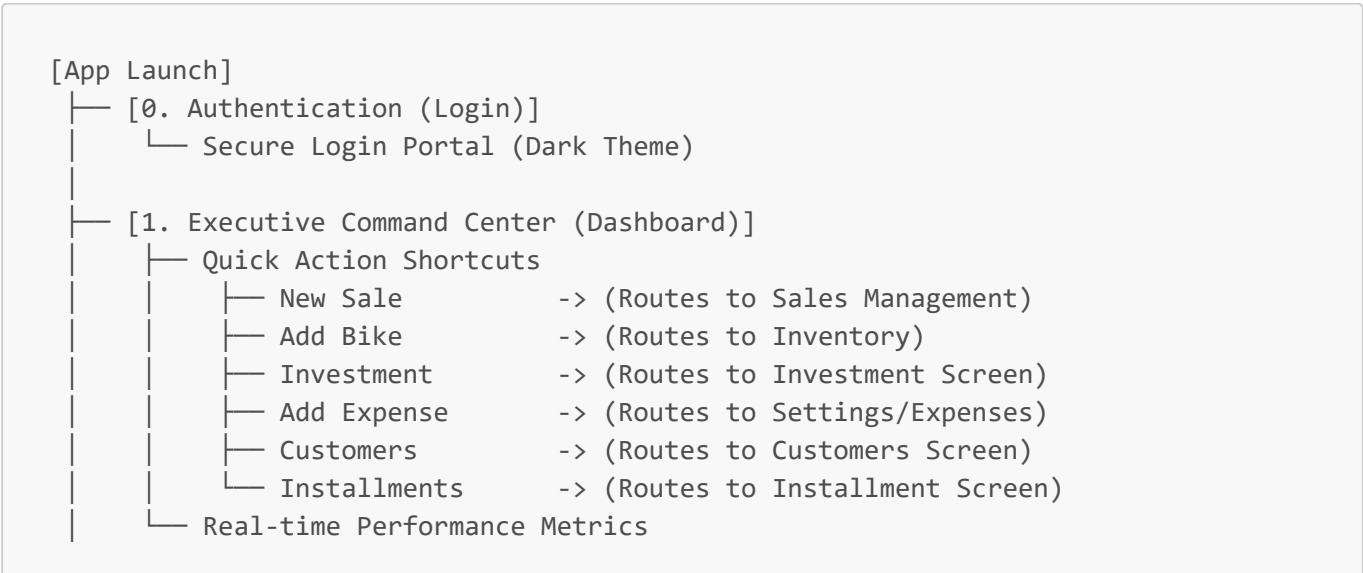
Role-Based Access Control (RBAC) & Security

The system secures critical paths using access control logic.

- **Administrative Rights:** Settings (Database Import/Export), Financial Configurations, and Capital Investments are locked behind secure administrative access.
- **Data Integrity:** Strict input validation ensures correct formatting (e.g., CNIC uploads, required pricing parameters) before any database commits occur.

2. Exhaustive Application User Flow (Tree Structure)

This tree represents the complete A-to-Z navigation architecture of the application.

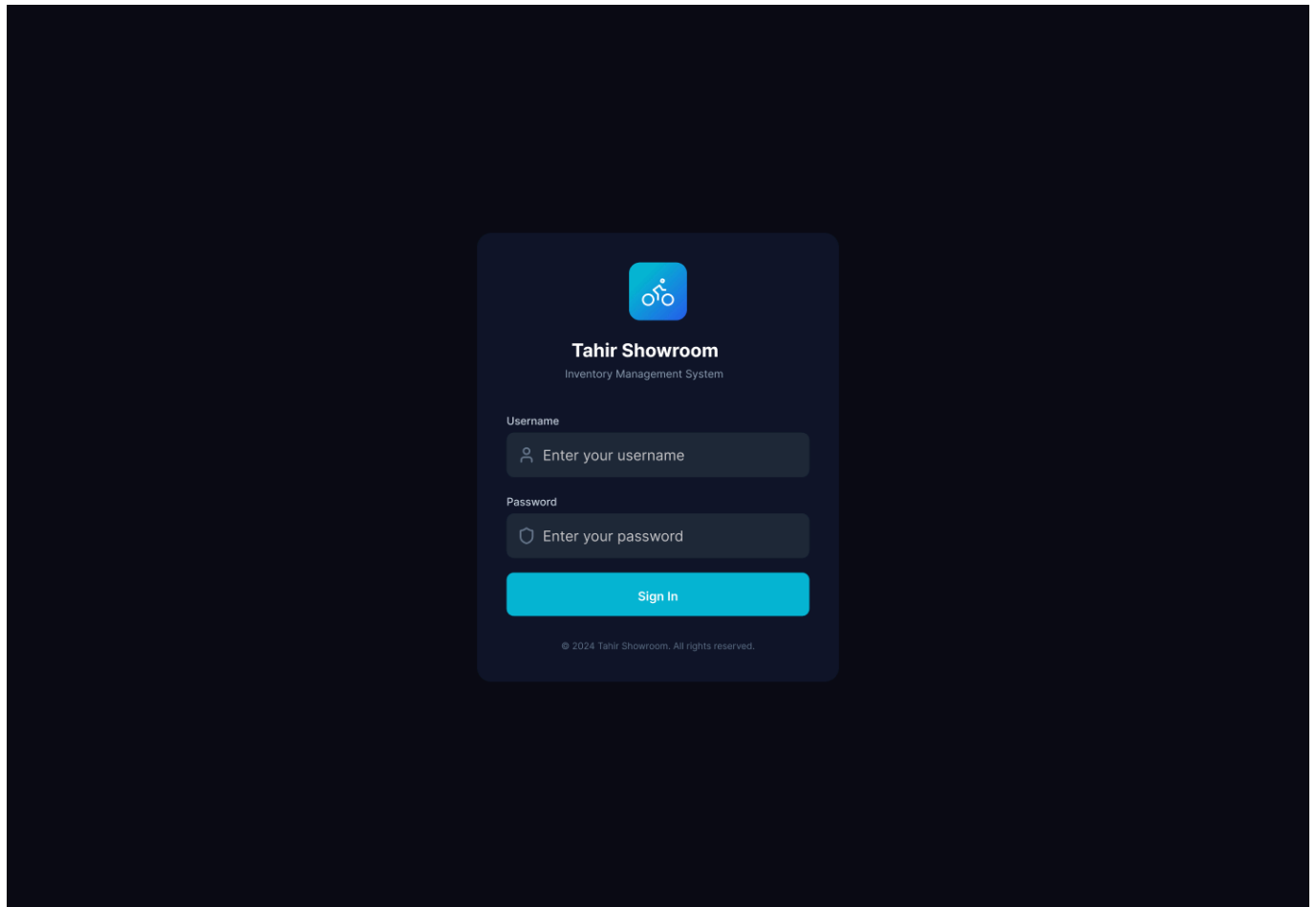


- └─ Capital Investment Details
 - └─ Units in Stock Detail & Low Stock Alerts
 - └─ Monthly Sales Revenue & Targets
 - └─ Total Installment Value & Contracts
 - └─ Performance Velocity (Weekly Sales Graph)
 - └─ Stock Allocation Distribution (Pie Chart)
- ─ [2. Dealers & Purchases Screen]
 - └─ Search Dealers (Search Bar)
 - └─ Add New Supplier
 - └─ Edit / Delete Supplier
 - └─ Add Stock Batch
 - └─ Supplier & Bike Details Inputs
 - └─ Total Cost of Bike(s)
 - └─ Add Multiple Bikes Feature
 - └─ Download Data Feature
- ─ [3. Inventory Management Screen]
 - └─ Filter & Search System
 - └─ Edit / Delete Bike Details
 - └─ Set Price Features
 - └─ Add Bike Screen
 - └─ Basic & Purchase Details
 - └─ Technical Specs
 - └─ Bike Papers Details
 - └─ Bike Image Upload
- ─ [4. Sales Management Screen]
 - └─ Filters & Search System
 - └─ Download / Delete Sales Data
 - └─ New Sales Screen
 - └─ Vehicle Selection
 - └─ Customer Information
 - └─ Witness Information
 - └─ Payment Terms & Contract Details
 - └─ Document Tracking
- ─ [5. Installments Screen]
 - └─ Filters & Search System
 - └─ Installment Metrics (Remaining days/months, Cleared payments)
 - └─ Download Installment Data Feature
- ─ [6. Customers Screen]
 - └─ Total Customers Metric & Data View
 - └─ Pending Payments & Active Installment History
 - └─ Edit, Delete & Download Customers
 - └─ Add New Customer Screen
 - └─ Personal & Contact Information
 - └─ Media Uploads (CNIC Photo & Profile Photo)
- ─ [7. Report Analysis Screen]
 - └─ Report Summaries & Download (Filter by Year/Month)
 - └─ Core Financial Displays (Available Cash, Net Profit, Total Expenses)
 - └─ Specific Bike Metric Displays

- [8. Investment Screen]
 - Core Financials (Total Invested, Available Cash, Net Profit)
 - Asset Valuation Metrics
 - Sold & Complete Bike Purchasing Values
 - Active Inventory Purchasing Values
 - Maintenance Spent & Total Expenses
 - Installment Predictions (Future Payment, Future Profit)
 - Filters & Separate Detail Views (Withdrawals vs Capital)
 - Download Investment Data
 - Add Capital Screen
 - Category Selection (Loan, Partnership, Other)
 - Amount, Date & Optional Note
 - Withdraw Screen
 - Amount, Date & Optional Note
 - Reason (Personal use, maintenance, expenses)
 - Capital Source Deduction Selection
- [9. Settings Screen]
 - Financial Section (Markup, EMI rounding, Late fees/percentages, Expense categories)
 - Inventory Section (Add/Edit Brands, Models, Model Years)
 - Database Section (Storage path, Export/Import Backup, Compress Media, Reset App)
 - Profile Section (Owner Name/Pic, Login User/Password)
 - General Section (Showroom Info/Logo, Currency, Dark/Light Theme, App Walkthrough)
- [10. Exit Screen]
 - Confirmation to Exit Options

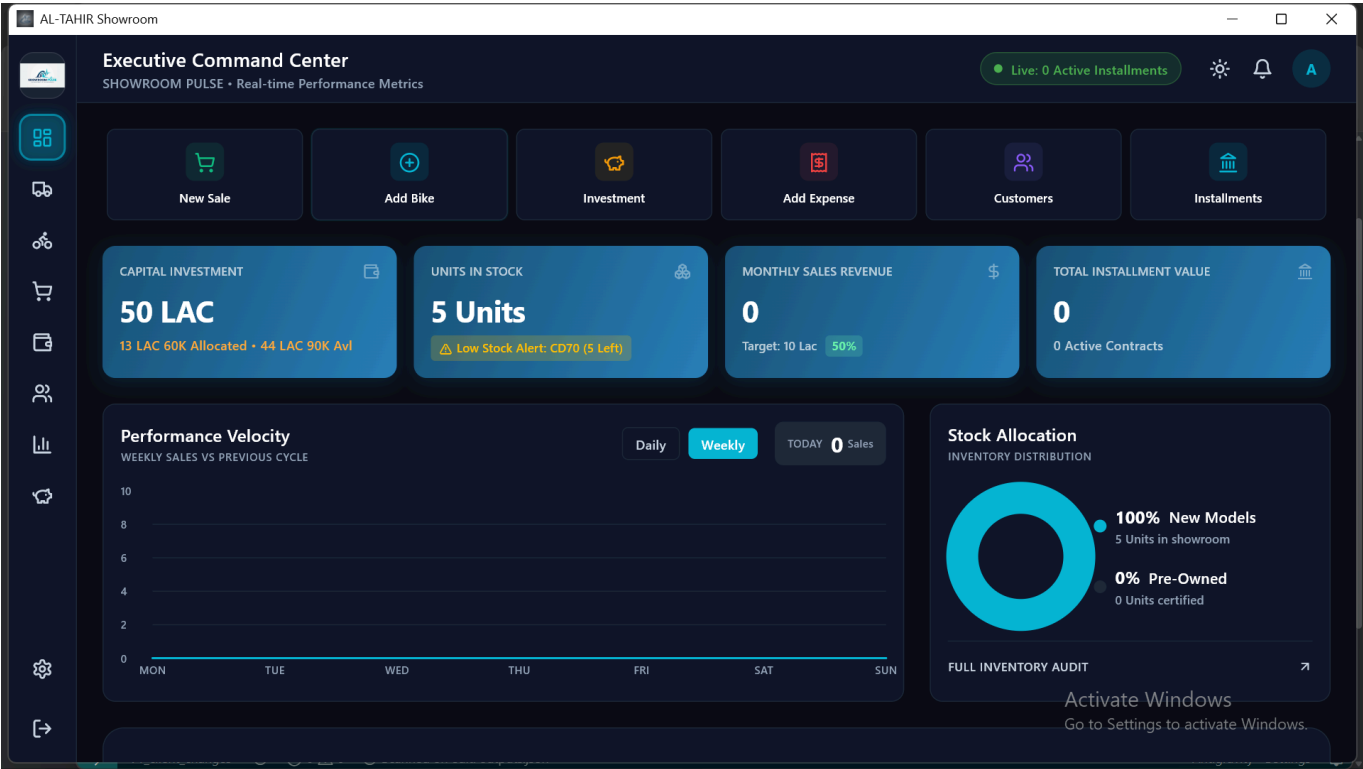
3. Visual Storyboard & Implementation Details

Page 1: Authentication Portal



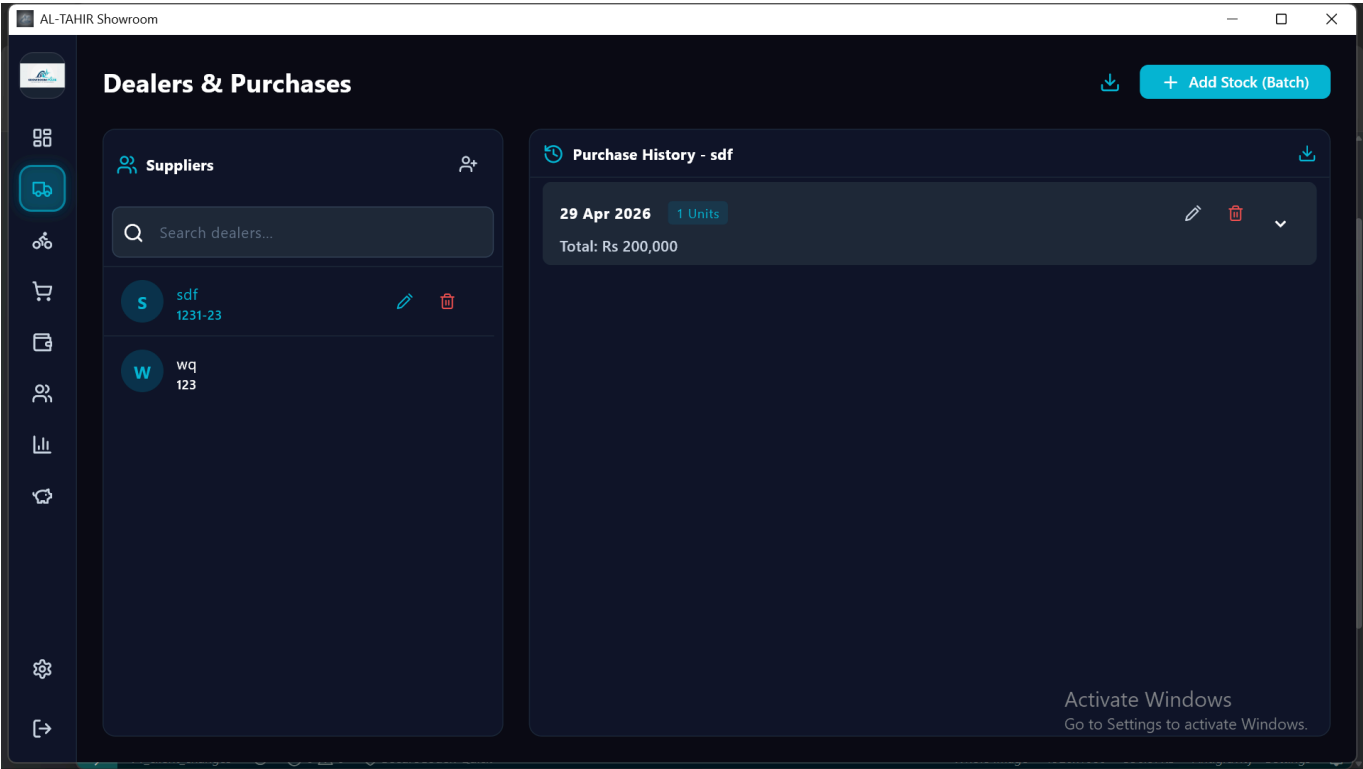
Caption: Secure entry point enforcing Role-Based Access Control (RBAC) and data isolation.

Page 2: The Executive Command Center



Caption: The central hub utilizing dynamic state management to render real-time KPIs (Capital, Sales, Stock) with quick-access routing.

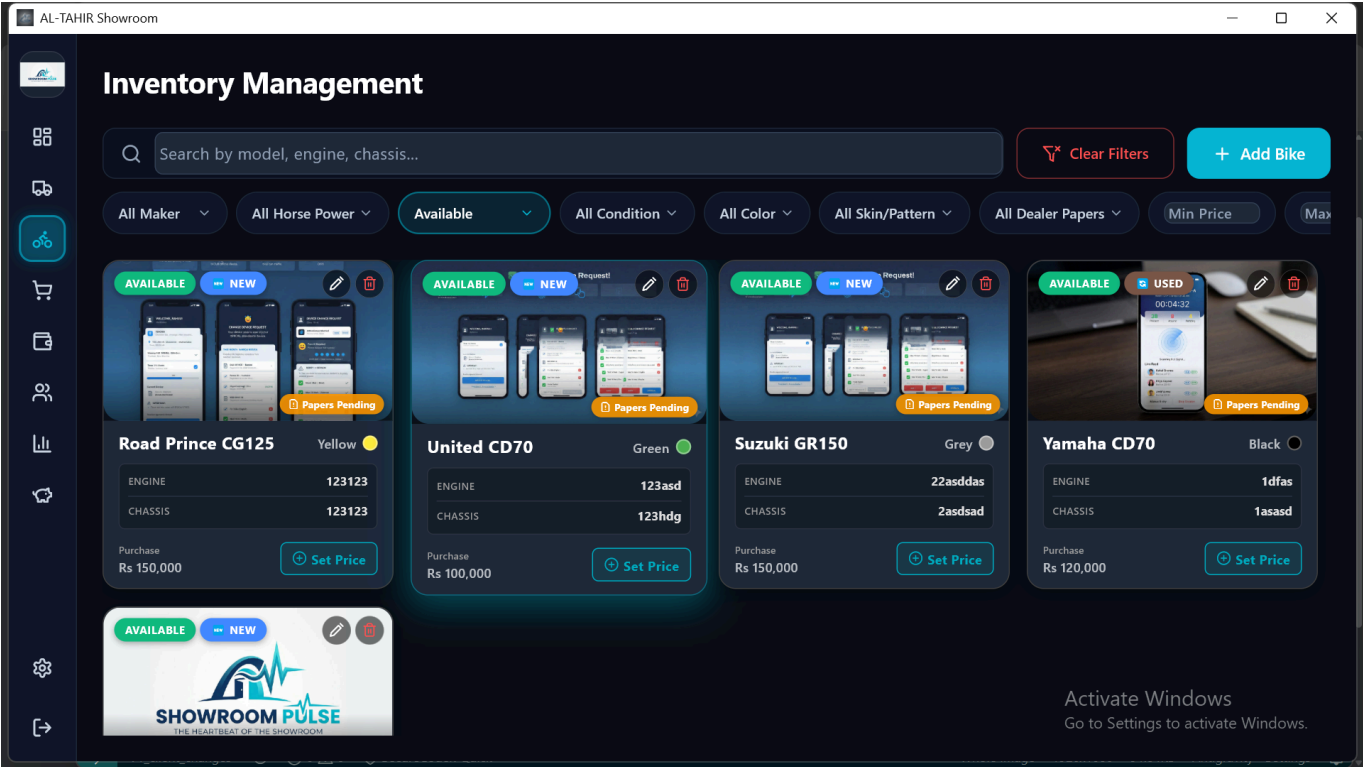
Page 3: Dealers & Purchases Management



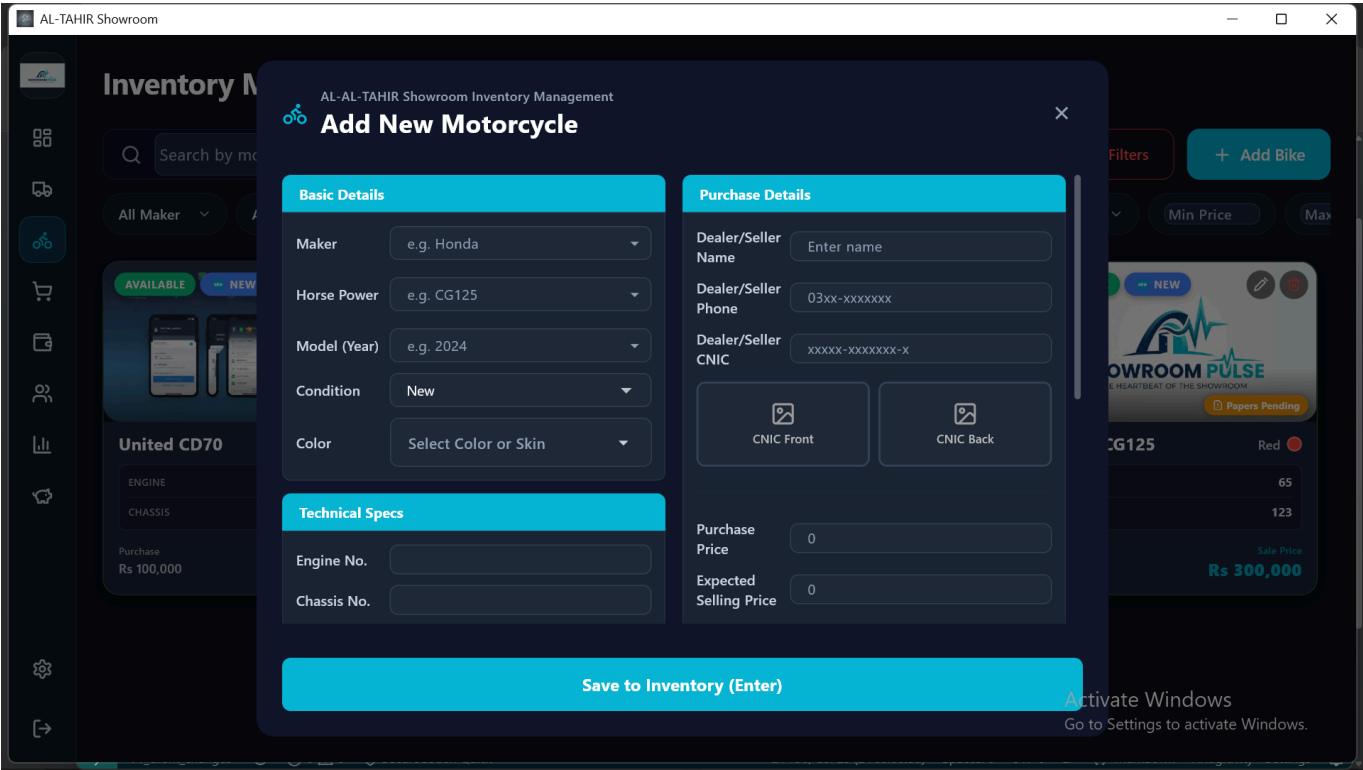
Caption: Supplier and procurement portal. Highlights the "Add Stock Batch" system calculating multi-unit costs.

Page 4: Inventory Engine

Inventory Management View:



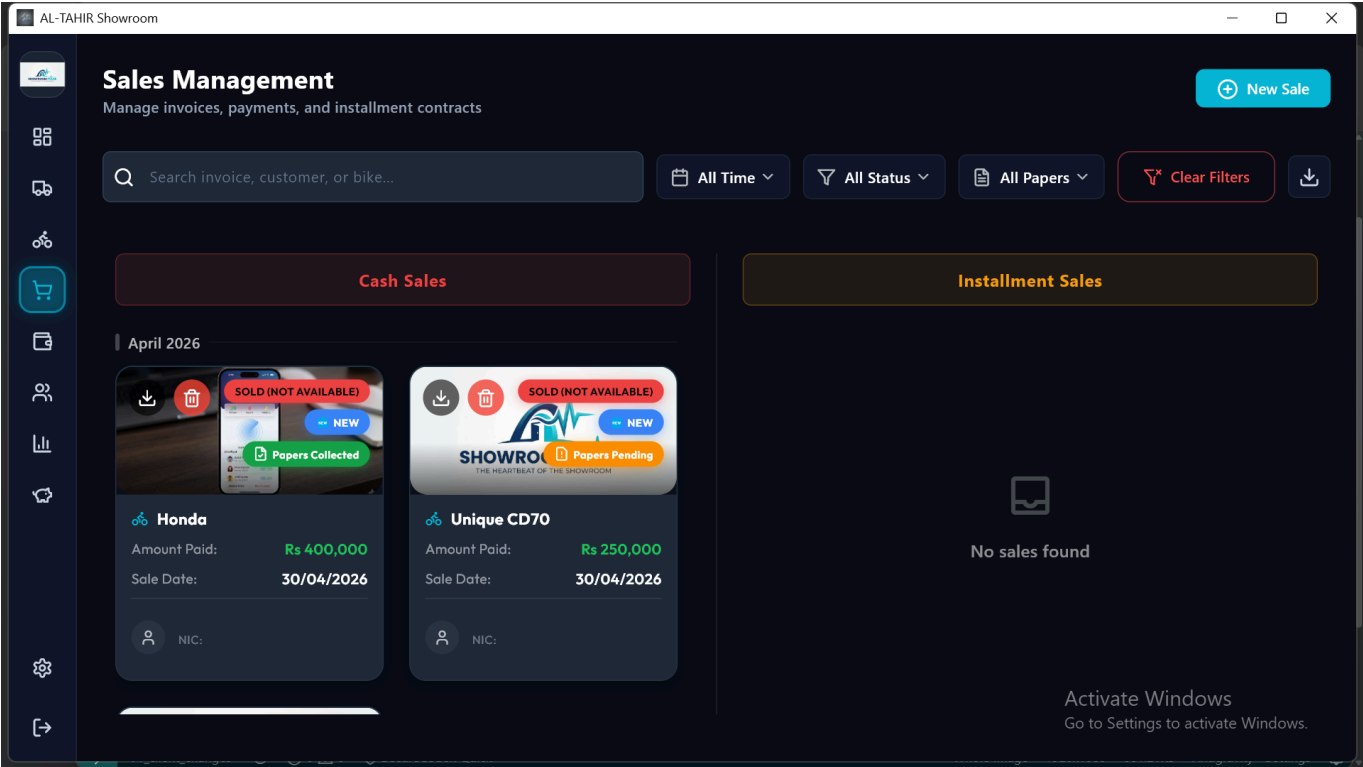
Add Bike Details:



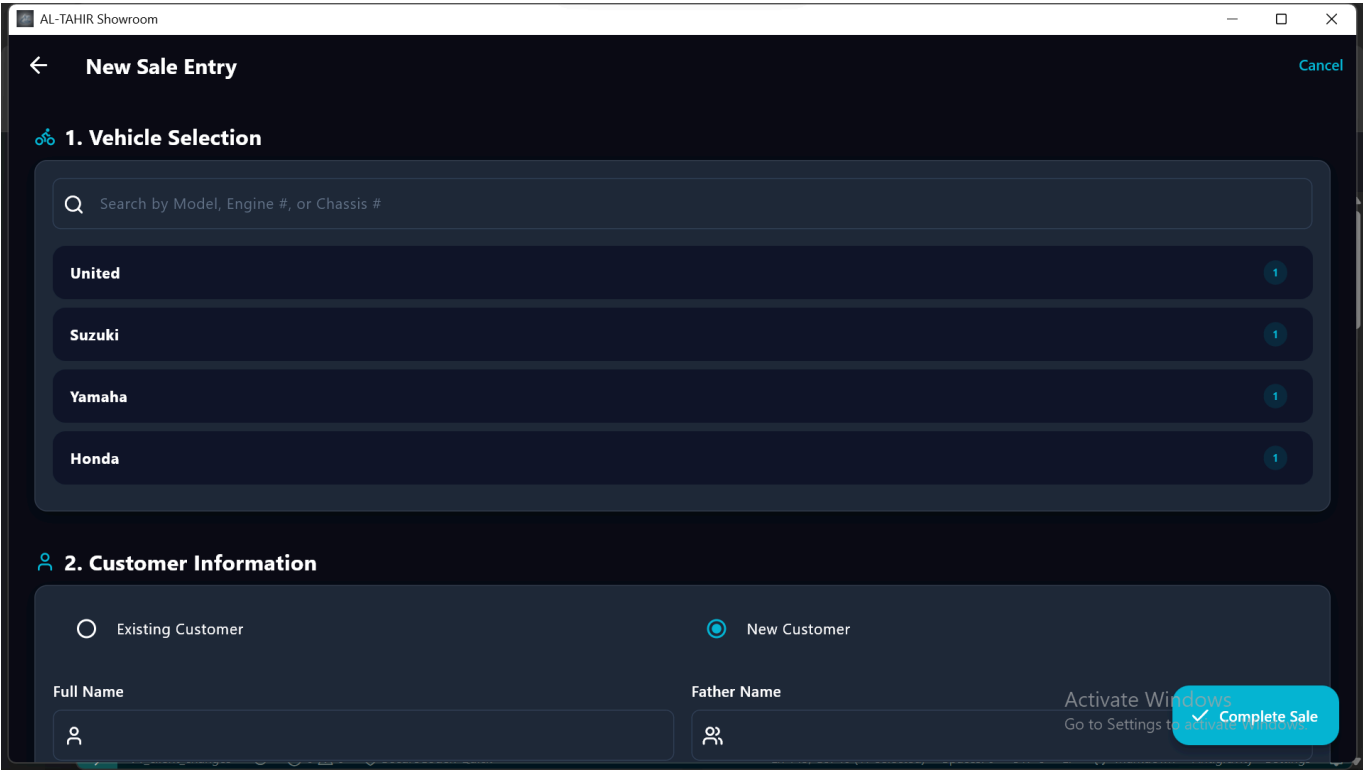
Caption: Comprehensive asset tracking with detailed specifications, pricing configurations, and image handling.

Page 5: Sales Orchestration

Sales Control Panel:



New Sale Generation:



Caption: End-to-end checkout system mapping vehicles to customers, enforcing contract terms, and handling document tracking.

Page 6: Installment Tracking

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Track Installments

Search by name or CNIC...

Due This Week

All Status

All Time

Clear Filters

Export All

?

CNIC:

Active

0/12 months

Road Prince

?

CNIC:

Horse Power: Road Prince

Chassis: 123123

Discount Added: Rs 100,000

Total Amount

Rs 450,000

Paid

Rs 25,000

Down Payment

Rs 25,000

Remaining

Rs 425,000

Next Due

30 Jun, 2026

in 29 days

Monthly EMI

Rs 35,400

Complete Installment Manually

Payment History

Record Payment

Date	Amount	Method	Collector	Status
31 May 2026	Rs 25,000	Cash	Online	Down Payment

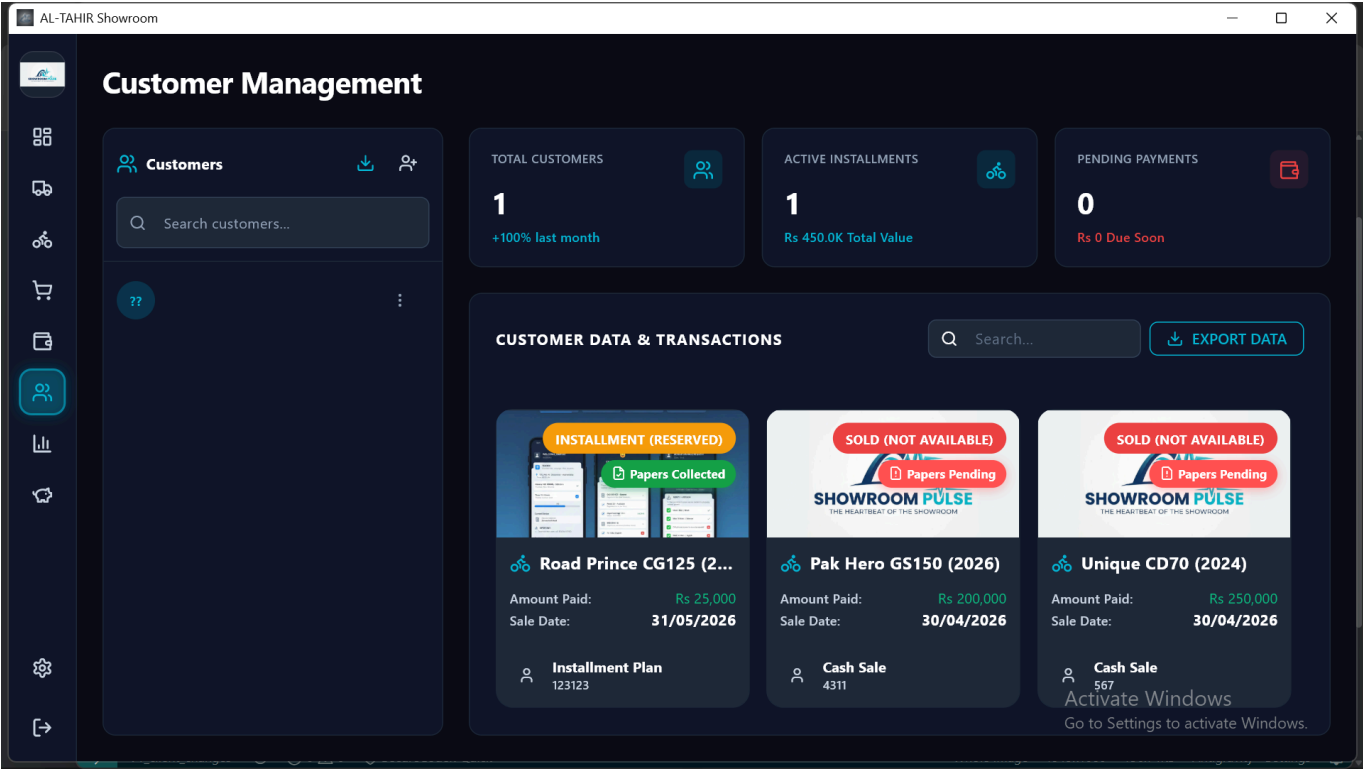
Activate Windows

Go to Settings to activate Windows.

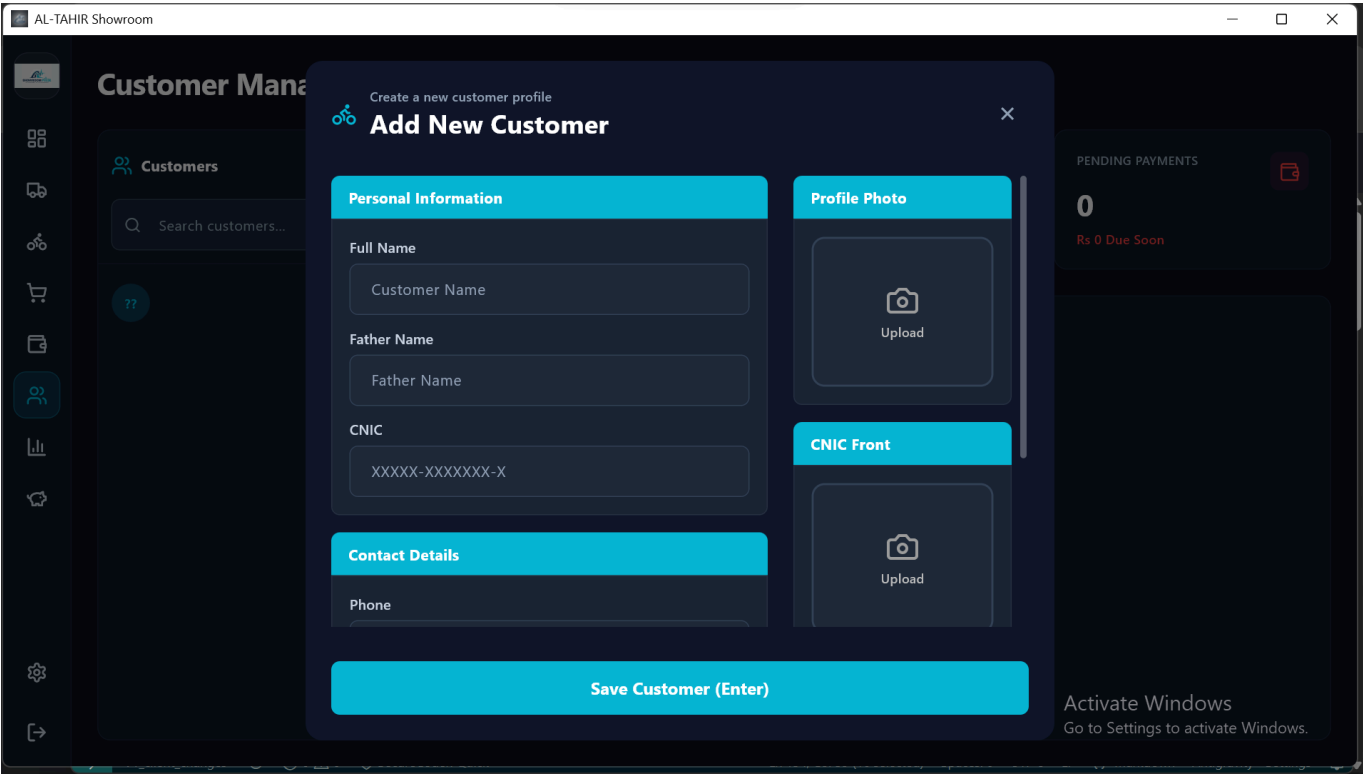
Caption: Automated tracking of remaining terms and cleared payments, heavily integrated with Sales data.

Page 7: Customer Relationship Management (CRM)

Customer Database:

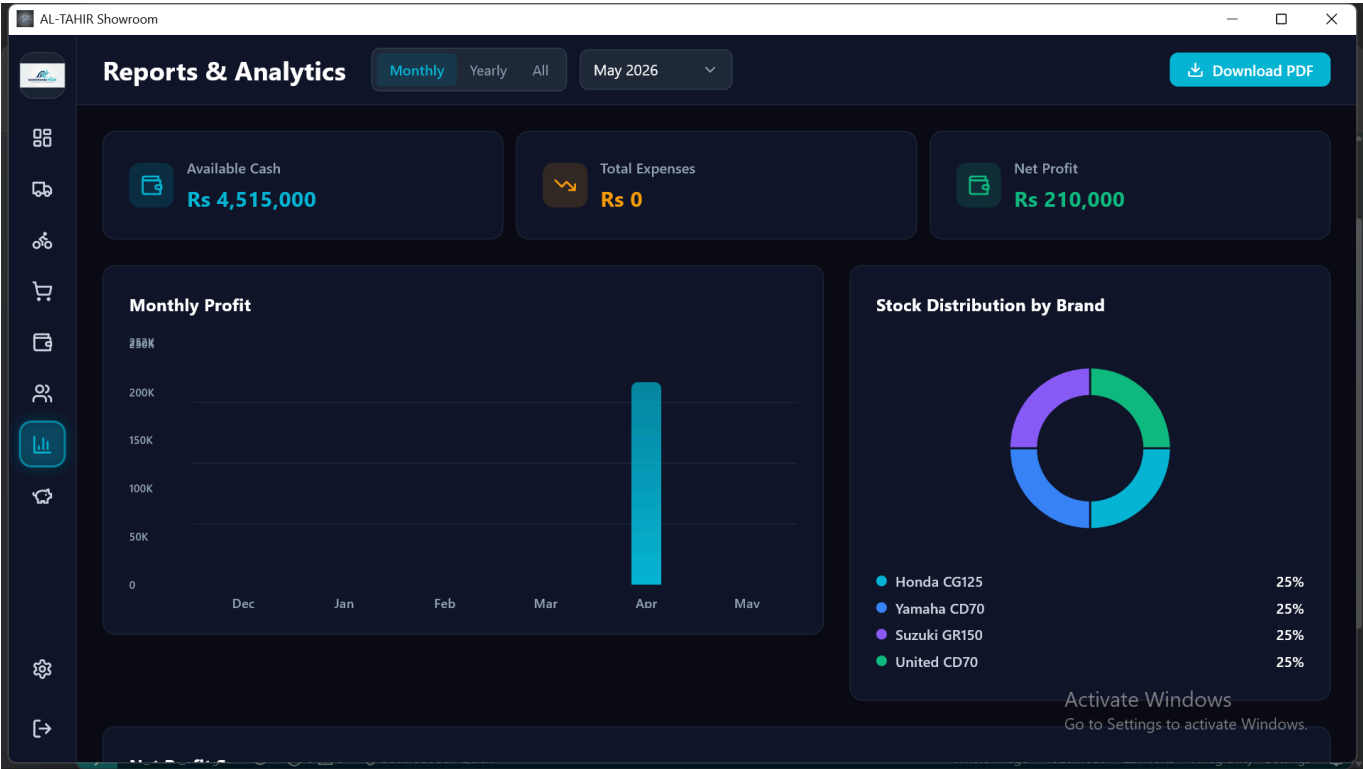


Add Customer Portal:



Caption: Secure profile management with integrated media upload (CNIC/Photos) and linked active payment histories.

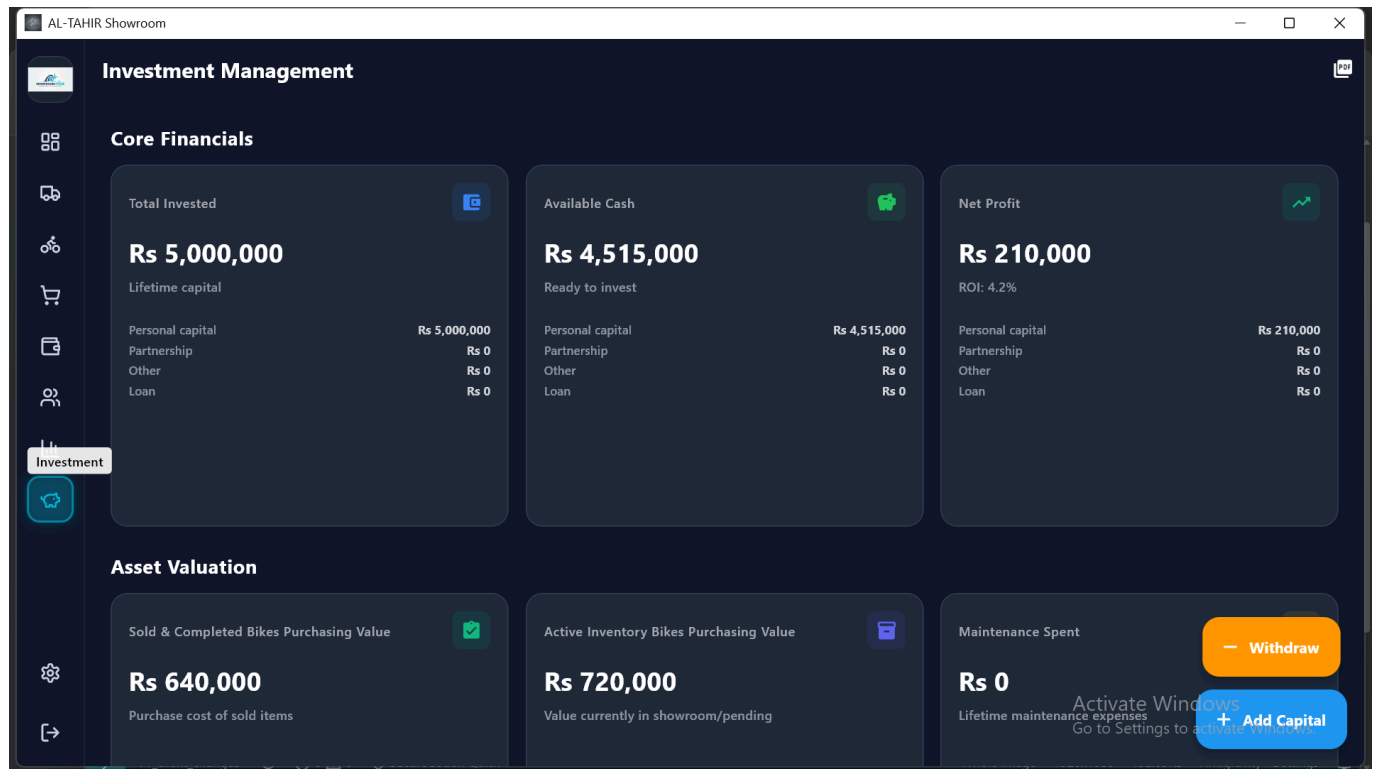
Page 8: Financial Report Analysis



Caption: Advanced financial aggregation summarizing available cash, net profits, and targeted time-based reporting.

Page 9: Investment & Asset Valuation

Asset Portfolio:

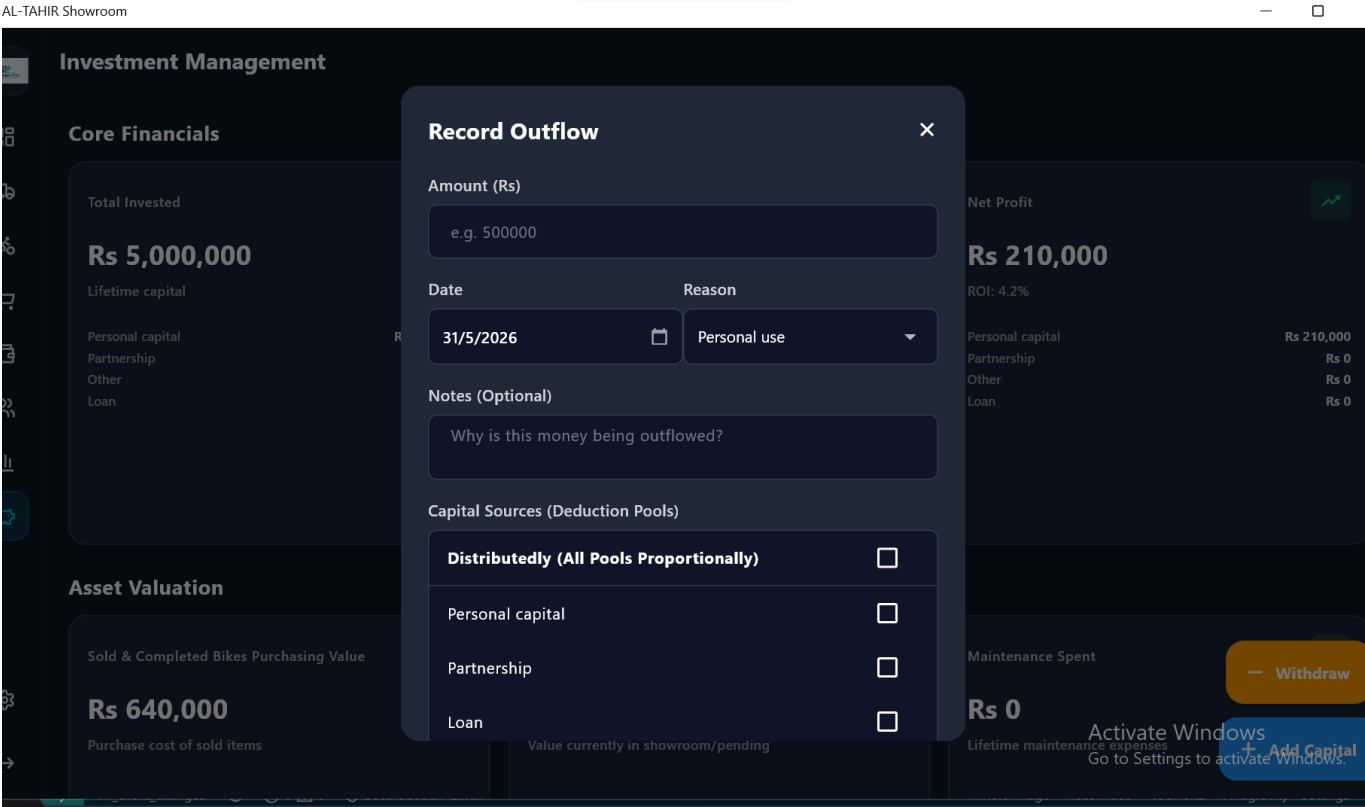


Transaction Handlers:

This screenshot shows the same dashboard as above, but with the 'Add Capital Investment' modal open in the center. The modal contains the following fields:

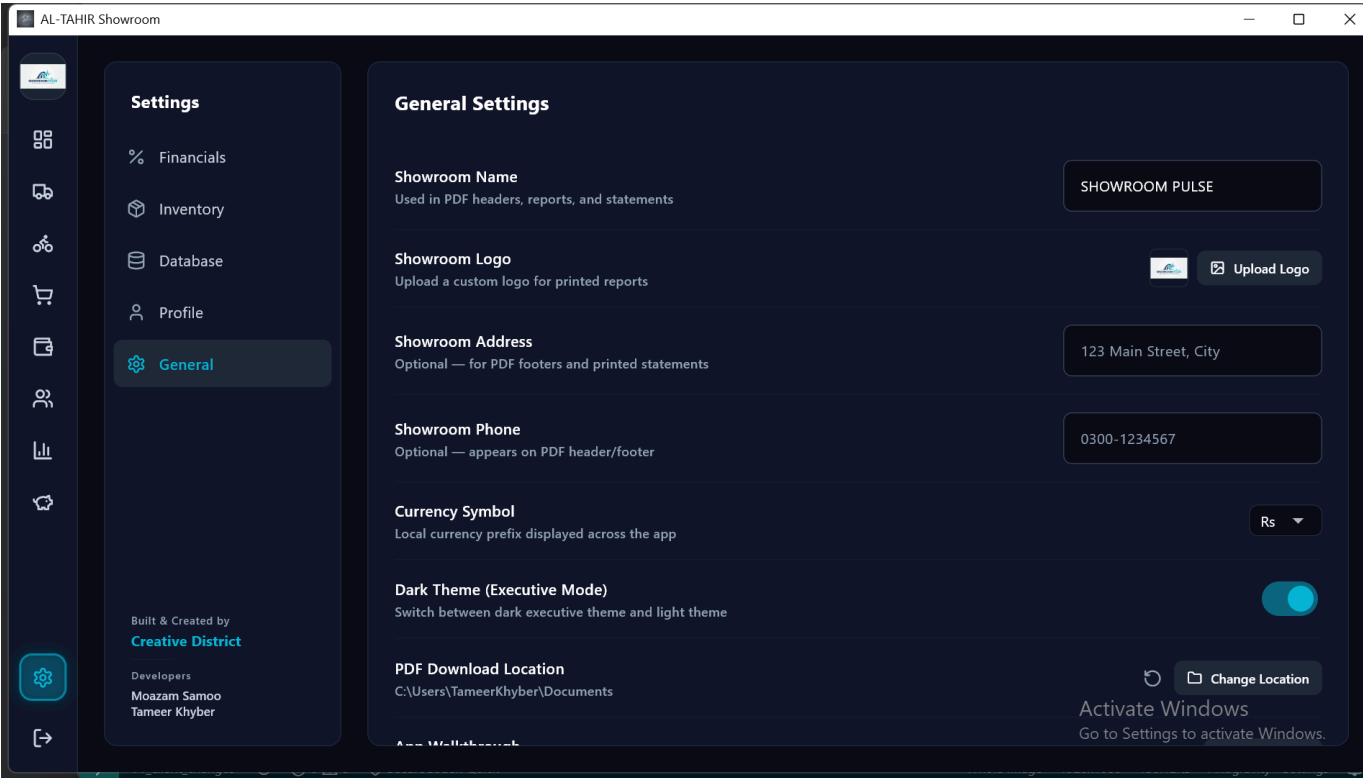
- Amount (Rs):** A text input field with the placeholder 'e.g. 500000'.
- Date:** A date picker showing '31/5/2026'.
- Category:** A dropdown menu currently set to 'Personal capital'.
- Notes (Optional):** A text area with the placeholder 'Why was this money invested?'.
- Save Investment:** A prominent blue button at the bottom of the modal.

The background dashboard content is partially obscured by the modal. The same watermark 'Activate Windows Go to Settings to activate Windows.' is present in the bottom right corner.



Caption: Core financial engine tracking investments versus withdrawals, providing predictive installment modeling and complete asset valuation.

Page 10: System Settings & Security



Caption: Administrative control center handling robust database backups, dynamic styling (Dark/Light mode), and core financial rules (EMI rounding, Markup).